Star Cores: book plan

Plan for DICTO's fifth book: spiked cores, their halos and the cycles they start



DICTO · plan of 30 October 2026 · for DICTO's decision · not for publication

1. What the book shows

DICTO's TI Star (#78) turned a solid's own pyramids outwards and found that the spikes start a cycle. Every such step, spike the faces and wrap the tips, is DICTO's growth rule T of Universal PiTet (#38) run backwards. So every solid has a star chain, and the book asks which chains close, how fast they grow and what cores they pass through.

> **Scope (DICTO, 30 October): every new shape gets the star core treatment.** Each of DICTO's new shapes with a model in the data set gets the treatment #78 gave TI: (1) its own pyramids from the centre; (2) the spikes, mirrored (×2) and golden (×φ); (3) the star, its hull of tips (the halo) and the exact fills, with overlaps checked; (4) the core of the turn, one solid per family of face planes, intersected; (5) the core's own turn (does it return?); (6) the star chain. The result is one card per shape in the book's catalogue.

> **Main statement (the book's aim, to be proved in Part III).** A solid whose faces lie in just two families of planes, on the five-fold and three-fold axes (or the four-fold and three-fold), has a star cycle in which the cores alternate with period two and the hulls with period four, as the truncated icosahedron does (#78, Propositions 4 and 5). Solids with more kinds of face plane run in longer chains, some of which pass through DICTO's named solids.

2. Title

> **Pick: *Star Cores: Spikes, Halos and the Cycles of DICTO's Solids*.** The words are your own: star core, Star Halo and TI Star.

3. Contents

Part	Chapter	Source	State
I. The rule	1. The Cleo star: the first spiked core	#35	**published**
	2. Universal PiTet: DICTO's growth rule T	#38	**published**
	3. One turn is a gnomon	#54	**published**
II. The TI Star	4. The Truncated Icosahedron Star (cycles, polar cycle, star chain, the core's own cycle)	#78 v2	**published**
III. New papers	5. Star Cores of DICTO's Solids: every new shape given the star core treatment, with its card	new paper A	**survey v2 run; exact work to do**
	6. Core Cycles: when the cores alternate (the main statement, proved)	new paper B	**to study**
	7. The Gains: the exact growth of every closed chain, and its primes	new paper C	**to study**
End	Catalogue of star cores, one card per new shape; table of every chain; index; the solids with D-numbers	book	**at assembly**

4. What is already known (survey v2, 55 solids of the data set)

- **Period four**, like TI: the GTI, the FPD and the Palintriacontahedron (one orbit, #38), Icosaxes, Unborn Axes, the Metaxyhexecontahedron, Tweedledum and Samson.
- **Period two:** David and Delilah.
- **Long chains through named solids:** TQHD → TQP → TQH → TPH → OG (#39), and OG → KRE → PHE → TPP → DPHP, after which the chain leaves the data set.
- **Not closed in eight steps:** 45 solids, among them the Queens' slices (Alice and the others). These are screened only: the step is valid only where the star lies inside its hull.
- **Found on the way:** the FPD (#38) and the Palintriacontahedron have the same shape.

5. Studies to run (all on dicto-big-brother, in this order)

#	Study	For
0	**The star core treatment for every new shape**, steps (1)–(6) above, for both ×2 and ×φ: the convex shapes first, then the non-convex ones from their centre where each face can see it (a shape that can't is reported, not forced). One card per shape: does the star fill, its halo, its core, whether the core returns, and the period of its chain.	A, B
1	Exact gains in $\mathbb{Q}(\sqrt{5})$ ($\mathbb{Q}(\sqrt{2})$ where needed) for the eight period-four and two period-two solids; check every step's fill and overlaps	A, C
2	The core of each turn (as Proposition 5 of #78) for the same ten solids: do the cores alternate with period two?	B
3	Proof of the main statement for two plane families (icosahedral and octahedral), and a search for counter-examples	B
4	The long chains: where DPHP's chain goes; whether OG's chain ever closes	A
5	The gains side by side, with their primes (TI: 89 and 109; ID: $10/\varphi^4$), to test DICTO's idea of a new series of ratios	C
6	Prior art: library and web search, every source listed	A, B, C

6. Decisions

- **FPD and Palintriacontahedron, one shape.** Pick: keep **FPD** (#38, the first name), with "Palintriacontahedron" noted as its other name. No D-number changes.
- **Overlap with The Golden Cycle.** #54 and #78 are in both books. Pick: keep both, since a book should read whole.
- **Order of work.** Pick: the studies first, then papers A, B and C, then the book. Each paper is released when ready, and the book comes after paper C.
- **An app feature for every paper.** Pick: a "Star step" button in Polyhedraverse that spikes any convex solid and wraps it, one tap per step. It is the View-in-app target for A, B and C.

Sources for this plan: #35, #38, #39, #54, #78 v2; studies 2026-10-30-star-chain-census (survey2.json) and 2026-10-30-core-cycle.